The Invisible Gap: Evaluating LLMs the Way Indians Actually Use Them

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Abstract

Romanized Latin script is widely used by speakers of Indic languages, yet open-ended generation in this representation remains poorly evaluated. We benchmark Claude, GPT-4o, Gemini, and Qwen3.5-35B on converting 142 English query–response pairs into romanized Hindi, Telugu, and Kannada. Outputs are evaluated using multilingual BERTScore and BGE-M3 cosine similarity under a failure-aware policy that assigns zero to empty outputs, English copies, native-script leakage, and repetition loops. GPT-4o and Claude achieve the strongest response-level BGE-M3 scores, ranging from 0.733 to 0.837 across the three languages. However, aggregate semantic scores obscure distinct failure modes. Gemini exhibits script leakage, affecting 24 Telugu and 19 Kannada responses, while Qwen frequently returns English unchanged and produces repetition loops. Verbatim English outputs can receive perfect similarity scores before failure-aware correction, demonstrating that semantic similarity alone does not measure successful romanization. Performance is consistently strongest for Hindi and weaker for Telugu and Kannada. These findings show that romanized Indic generation should be evaluated using semantic similarity together with completion reliability, script compliance, English-copy detection, and repetition analysis.

1 Introduction

For many speakers of Indian languages, communicating online does not require switching to a native-script keyboard. A Hindi speaker may type aap kaise hain, a Telugu speaker meeru ela unnaru, and a Kannada speaker neevu hegiddira. Although these sentences use Latin characters, they are not English. Their vocabulary, grammar, meaning, and social register belong to the corresponding native language. Romanized text is therefore not merely a transliteration convenience; for many users, it is the practical interface through which they communicate with digital systems, including large language models (LLMs).

This behavior is common in real-world communication. Romanized text accounts for 83% of Hindi and 73% of Telugu messages in a corpus of patient WhatsApp interactions Khullar et al. (2025). However, most multilingual NLP benchmarks evaluate languages in their standardized native scripts. Resources such as XQuAD Artetxe et al. (2020) and IndicQA Doddapaneni et al. (2023) contain formal native-script text, while datasets such as Dakshina Roark et al. (2020) and Aksharantar Madhani et al. (2023) primarily study transliteration. Consequently, the ability of an LLM to generate long, natural, and semantically faithful Romanized Indic text remains insufficiently understood.

At first glance, this task may appear to be ordinary translation followed by transliteration. In practice, it requires several capabilities at once. The model must understand the English source, express the same meaning naturally in the target language, write that language entirely in Latin characters, preserve names and technical terms, and maintain an appropriate register. Romanization also lacks a universally accepted spelling standard. A model can therefore produce text that is superficially plausible while using unnatural vocabulary, English sentence structure, inconsistent spellings, or characters from the wrong script.

We formulate the task as semantic alignment between an English source and its Romanized Indic counterpart. A successful output should preserve the meaning of the English text without simply copying it. It should also satisfy the representation constraint: the result must be Hindi, Telugu, or Kannada written in Latin script. This distinction is important because a verbatim English response has perfect semantic similarity to the English reference, but it is a complete failure of the requested Romanized-language generation task. Similarly, an answer written in native script may preserve meaning while violating the interface expected by the user.

Our initial experiments revealed that these failures become more pronounced for long responses. Short queries were often converted successfully, while longer answers were more likely to remain in English, contain native-script characters, omit content, produce incomplete structured output, or collapse into repetition. This motivates evaluating the query and response separately, while still generating them together.

We use one model call for each paired query and response. The model receives both English fields and returns romanized_query and romanized_response in a single structured output. This design preserves the conversational relationship between the user request and its answer. In pilot experiments, generating the fields independently caused the model to lose context and behave inconsistently, particularly when processing long responses. Joint generation allows the model to interpret the response in light of the query while still permitting separate evaluation of both fields.

We evaluate four multilingual LLMs: Claude, GPT-4o, Gemini, and the locally served Qwen3.5-35B model. Each model converts the same 142 English query–response pairs from OpenAssistant Köpf et al. (2023) into Romanized Hindi, Telugu, and Kannada. This controlled design gives every model the same source content and supports direct comparison across models and languages.

Our evaluation considers three complementary properties. First, we measure semantic alignment using multilingual BERTScore Zhang et al. (2020) and BGE-M3 cosine similarity Chen et al. (2024). Second, we measure generation reliability by detecting empty fields, English copies, native-script leakage, incomplete outputs, and repetition loops. Third, we examine generation efficiency through output-token usage and completion behavior. Efficiency matters because a model that requires substantially more tokens, repeatedly reaches its generation limit, or fails to complete long responses may be impractical even when its successful outputs preserve meaning.

The results show that multilingual capability does not automatically imply reliable Romanized-language generation. Hindi is consistently the strongest language across models, while Telugu and Kannada expose larger gaps. Claude and GPT-4o obtain the highest response-level BGE-M3 scores, ranging from 0.733 to 0.837 across the three languages. Nevertheless, aggregate similarity scores hide important failure modes. Gemini frequently leaks non-Latin characters, including native Indic and unrelated scripts. Script leakage affects 24 Telugu and 19 Kannada responses. Qwen exhibits a different failure pattern: it often returns the English source unchanged for Telugu and Kannada, and it is the only evaluated model to produce severe repetition loops.

These observations also expose a limitation of standard semantic metrics. Before failure-aware correction, a verbatim English output receives a BGE-M3 cosine similarity and BERTScore of 1.0 because it exactly matches the English reference. The metric therefore gives its highest possible score to an output that did not perform the requested task. To address this problem, we introduce a failure-aware policy in which English copies, empty or unusable fields, native-script corruption, and detected repetition loops receive zero. Unlike evaluation that silently removes unsuccessful generations, this policy keeps all 142 samples in the denominator and measures semantic quality together with task completion.

Our contributions are:

- We present a controlled benchmark for paired English-to-Romanized Hindi, Telugu, and Kannada generation across four multilingual LLMs.

- We evaluate semantic alignment separately for queries and responses while generating both fields jointly to preserve conversational context.
- We introduce a failure-aware evaluation policy that prevents English copies, empty outputs, native-script leakage, and repetition loops from receiving misleadingly high aggregate scores.
- We identify distinct model-specific failure modes: script instability in Gemini and English copying and degeneration in Qwen, especially for Telugu and Kannada.
- We examine generation efficiency through output-token usage, completion reliability, and behavior on long responses.

2 Related Work

The starting point for this work is a tokenization inequality. Petrov et al. Petrov et al. (2023) showed that tokenizers trained on English-dominant corpora require up to 15× more tokens to represent the same content in non-English languages. For a Telugu speaker, this means every interaction with a commercial LLM is structurally more expensive—in cost, latency, and effective context window—before the model processes a single word. That asymmetry raises a deeper question: if the infrastructure already disadvantages non-English users, how well do LLMs actually serve them in practice?

For hundreds of millions of Indians, the practical answer involves romanized script. Rather than switching to a native-script keyboard, Telugu, Hindi, and Kannada speakers type their language in Latin characters—a behavior documented at 83% prevalence for Hindi and 73% for Telugu in real LLM deployments Khullar et al. (2025). This romanization is not standardized: the same word may be spelled a dozen ways depending on the speaker, a variability documented across South Asian languages in the Dakshina Roark et al. (2020) and Aksharantar Madhani et al. (2023) corpora.

Prior work has begun to characterize what this means for LLM performance. Khullar et al. Khullar et al. (2025) showed that romanized input degrades LLM performance on structured triage classification by up to 24 points relative to native script. Husain et al. Husain et al. (2024) took the complementary view—using romanization as an input strategy to unlock multilingual capabilities—and found it improves performance on formal benchmarks. Both studies, however, evaluate against tasks with a known correct answer. Neither asks whether automatic metrics can detect quality failures on open-ended generation tasks, where no ground truth exists.

Standard evaluation benchmarks such as XQuAD Artetxe et al. (2020), IndicQA Doddapaneni et al. (2023), and MLQA use formal native script and presuppose a reference answer to score against. Multilingual BERTScore Zhang et al. (2020) and BGE-M3 cosine similarity Chen et al. (2024) extend this paradigm to embedding space—but they still compare outputs to a reference, with no mechanism to detect whether a response uses the wrong register, chooses unnatural vocabulary, or reads like a translation rather than something a native speaker would write. We fill this gap: using native-speaker human evaluation as ground truth, we show that these metrics produce scores in a narrow, acceptable-looking band even when the outputs they are rating are rejected by the very speakers the system is supposed to serve.

3 Method

3.1 Data

We use 142 English query–response pairs derived from a processed subset of the OpenAssistant dataset (OASST2) Köpf et al. (2023). Each record contains a message identifier, a high-quality English query, and a corresponding English response. The same 142 records are used for all three languages and all four models, enabling direct cross-model and cross-language comparison under identical input conditions.

3.2 Models

We selected a hybrid of closed-source and open-weight models to compare proprietary API-based systems with a locally deployable configuration. The closed-source models are Claude Opus 4.8, accessed through the Anthropic API; GPT-4o, accessed through the OpenAI API; and Gemini, accessed through the Google Generative AI API. All three models receive the same structured prompt and are instructed to return a JSON object containing romanized_query and romanized_response. For the open-weight configuration, we serve the quantized qwen3.5:35b model locally through Ollama on an NVIDIA RTX PRO 6000 Blackwell GPU. We disable model thinking, use temperature zero, and permit up to 4096 generated tokens with a 300-second timeout for the qwen.

3.3 Generation

Each model receives the English query and response together in a single request and is instructed to return a JSON object with two fields: romanized_query and romanized_response. Joint generation preserves the semantic relationship between query and response and avoids the context loss observed when the two fields are generated independently.

The prompt specifies that output must use only Latin characters, that native script must not appear in the romanized text, and that natural code-mixing with English technical terms is acceptable. For Kannada, we use a stricter language-specific prompt that explicitly prohibits both native Kannada script and English structural borrowing in the final output, following pilot observations that Kannada generation required tighter constraints.

Malformed or partial outputs are retained when recoverable; failed fields remain blank.

3.4 Evaluation

Automated semantic evaluation. For each non-empty generated response, we compare the original English response against its romanized counterpart with two automatic metrics. BERTScore F1 is computed using bert-base-multilingual-cased with baseline rescaling disabled Zhang et al. (2020); we treat this as a secondary token-alignment signal. BGE-M3 cosine similarity is our primary cross-lingual semantic measure, computed by embedding both texts with the BGE-M3 model and taking the cosine of the resulting dense vectors Chen et al. (2024).

Failure-aware coverage policy. We define a row as invalid if any of the following conditions hold: (1) the romanized response is empty or whitespace-only; (2) the romanized response is identical to the whitespace-normalized English source (word overlap ratio = 1.0), indicating that the model copied the input rather than romanizing it; or (3) the romanized response contains one or more native-script characters, indicating script corruption; or (4) the romanized response is degenerate—a single token repeated at least 20 times consecutively, or a four-token phrase repeated at least 50 times—indicating a repetition loop. Invalid rows receive zero for all automatic metrics rather than being excluded from the dataset. This policy ensures that all 142 samples enter each aggregate mean, so reliability and semantic quality—including resistance to degeneration—are reported on the same denominator.

Human evaluation. Native speakers of Hindi, Telugu, and Kannada independently rated a purposively sampled set of approximately 24 responses per language (73 total). Rather than random sampling, we selected rows to surface the specific ways romanized output fails while remaining manageable for native-speaker review. Selection targeted six categories: (1) pure/mostly English—outputs returned in English where automatic metrics still score high; (2) native-script leakage—romanized text with stray script characters; (3) metric divergence—rows where BGE and BERTScore disagree by more than 0.15 points; (4) hard prompts—creative, emotional, or roleplay prompts where quality is difficult to auto-judge; (5) good anchors—clean high-quality responses included as a baseline; and (6) English-task rows—prompts that operate on English text (OCR cleanup, grammar correction, email improvement) where English output is defensible and flagged separately. Because rows are

Model	Language	BGE Cosine	BERT-F1	Invalid
Anthropic	Hindi	0.837	0.676	0
	Telugu	0.789	0.671	4
	Kannada	0.733	0.620	5
GPT-4o	Hindi	0.828	0.663	0
	Telugu	0.794	0.667	0
	Kannada	0.749	0.641	2
Gemini	Hindi	0.763	0.617	10
	Telugu	0.630	0.548	25
	Kannada	0.664	0.569	19
Ollama	Hindi	0.787	0.639	4
	Telugu	0.598	0.574	16
	Kannada	0.537	0.549	20

Table 1: Failure-aware automatic evaluation scores (response only). BGE Cosine is BGE-M3 cosine similarity between the English reference and the romanized output; BERT-F1 is raw multilingual BERTScore F1 using bert-base-multilingual-cased. Invalid counts rows that received zero (empty, English copy, or native-script corruption) out of 142 total.

selected on failure conditions, per-model counts within the diagnostic set do not represent population failure rates; those are computed separately over all 142 responses per model.

Evaluators scored each response on five dimensions: naturalness and fluency (0–3), grammatical correctness (0–2), word choice accuracy (0–2), register appropriateness (0–2), and overall usability (0–3), for a maximum of 12 points per response. To accelerate review, we pre-filled scoring sheets using Claude Opus 4.8 as an LLM judge; native speakers then read each romanized_response and corrected any value they disagreed with. The final H6–H10 scores are native-speaker judgments; the LLM draft served as a time-saving starting point.

4 Results

4.1 Automatic Evaluation

Table 1 reports failure-aware BGE-M3 cosine similarity and BERTScore F1 for all four models across the three languages. All 142 samples enter each mean; rows flagged as invalid (empty output, English copy, or native-script corruption) receive zero for all metrics rather than being excluded.

Language-level performance. Hindi obtains the highest scores across all four models (BGE: 0.763–0.837), followed by Telugu (0.598–0.794) and Kannada (0.537–0.749). This ordering is consistent with the availability of romanized Hindi training data: Hinglish is the most widely represented romanized Indic variety in publicly available corpora, while romanized Kannada and Telugu appear far less frequently.

Model-level performance. Anthropic and GPT-4o produce the strongest and most reliable outputs. Both achieve zero invalid rows for Hindi, and at most two for Telugu and Kannada. Their BGE scores for Hindi (0.828–0.837) are the highest observed. Gemini and Ollama degrade sharply on Telugu and Kannada, but for distinct reasons examined in Section 4.2.

Misleadingly acceptable score range. A notable pattern is the narrowness of the BGE range for the two strongest models: Anthropic and GPT-4o produce scores between 0.733 and 0.837 across all language conditions. Considered in isolation, this band appears to indicate acceptable semantic preservation. Section 4.3 shows that native speakers do not agree.

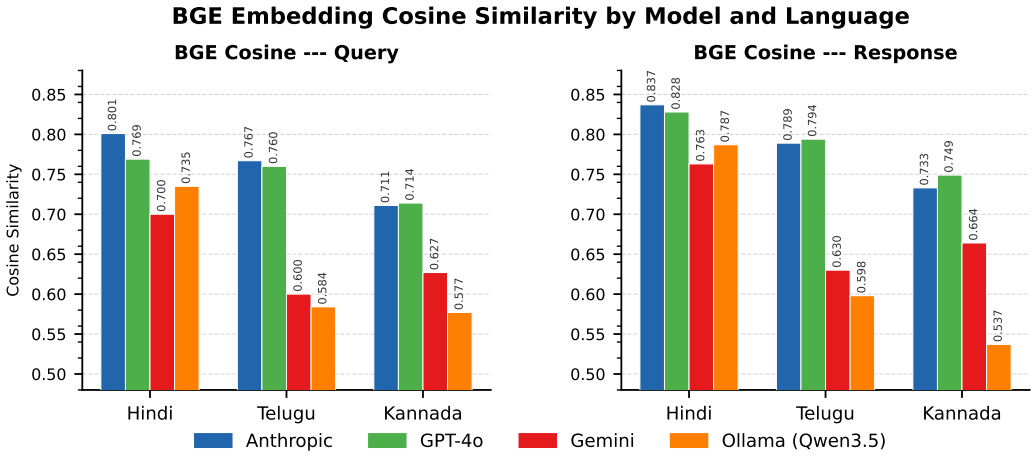


Figure 1: BGE-M3 cosine similarity for romanized queries (left) and responses (right) across all four models and three languages. Scores are failure-aware: empty, repeated, and script-corrupted outputs receive zero and are included in the mean over all 142 samples.

Model	Language	English-only	Script Spillage
Anthropic	Hindi	0	0
	Telugu	0	3
	Kannada	0	5
GPT-4o	Hindi	0	0
	Telugu	0	0
	Kannada	0	2
Gemini	Hindi	0	8
	Telugu	0	24
	Kannada	1	19
Ollama	Hindi	3	0
	Telugu	16	0
	Kannada	19	0

Table 2: Failure breakdown per model and language (out of 142 rows). English-only = response returned in English (word-overlap ≥ 0.90) instead of the romanized target language. Script Spillage = response contains one or more non-Latin characters (native script, Cyrillic, or other), after excluding legitimate romanization diacritics (ā, ē, ṭ, etc.) and Catalan source-title characters.

4.2 Failure Analysis

Table 2 breaks down the two primary failure modes: responses identical to the English source (repetition) and responses containing native script characters (script corruption).

Gemini and Ollama represent qualitatively different failure modes. Gemini produces script spillage: non-Latin characters appear mid-sentence in otherwise romanized output. A detailed character-level scan of all responses (after excluding legitimate romanization diacritics such as ā, ē, ṭ and a Catalan source-title character in one play prompt) reveals that spillage affects 24 of 142 Telugu responses (16.9%), 19 of 142 Kannada responses (13.4%), and 8 of 142 Hindi responses (5.6%) from Gemini. The Telugu case is the most severe: Gemini produces entire paragraphs in native Telugu script across 24 rows, totaling 2,635 Telugu characters. Anthropic shows minor spillage confined to Kannada (5 rows) and Telugu (3 rows). GPT-4o is nearly clean, with only 21 Kannada characters across 2 rows. Ollama produces zero script spillage across all three languages.

Closer inspection reveals two unexpected spillage sub-types. First, Cyrillic leakage: Gemini Telugu row 53 inserts the Russian word ("the only one") into a mid-sentence position in an otherwise Telugu-romanized narrative—an entirely non-Indic script appearing in a Telugu romanization task. Anthropic Kannada also shows isolated Cyrillic characters across 3 rows, unrelated to any Russian source content. Second, cross-language script confusion: Gemini's Kannada row 33 contains Telugu script ( ) in a response that should be Kannada; Gemini's Telugu row 19 simultaneously contains both Tamil script ( ) and Telugu script in the same line—two wrong scripts simultaneously. In Gemini Hindi, script spillage takes a subtler form: the Devanagari word ("believes") recurs across 8 rows, left in native script mid-Hinglish sentence rather than being romanized. These patterns suggest that Gemini's script boundary is unstable in a way that standard automatic metrics cannot detect: BGE-M3 and BERTScore measure semantic content, not whether the output characters are in the correct script.

Ollama's dominant failure is English-only output: the model returns the answer in English instead of the romanized target language. Scanning all ∼142 responses per model with a word-overlap threshold of $\geq$0.90 reveals 70 such instances in total. GPT-4o produces zero English-only outputs across all three languages. Anthropic (10) and Gemini (9) cases are almost entirely English-task rows—prompts that operate on English text (OCR cleanup, grammar correction, email improvement) where English output is defensible; only ∼2 per model are genuine failures on prompts that expected the target language. Ollama accounts for 51 instances, of which 41 are genuine failures on informational, creative, and conversational prompts. English-only rates reach 14.7% for Telugu and 30.4% for Kannada, while Hindi is largely unaffected at 2.8%.

A second Ollama-specific failure is degeneration: the model collapses into repeating a single word or short phrase for hundreds to thousands of tokens. We identify 13 such instances across the full dataset—every one produced by Ollama, zero by any other model. The most extreme case is a Kannada spelling-correction prompt where the word ellu repeats 2,017 consecutive times across 2,020 total words. The automatic consistency check (parameter P4 of our rubric) scores all 13 loops a perfect 3, and BGE-M3 assigns scores of 0.38–0.72—sometimes still "passing." Human raters assign usability zero to every degeneration case.

The two failure types do not overlap: Gemini never produces English-only output and Ollama never leaks native script, suggesting the failures originate in different aspects of instruction following. BERTScore and BGE-M3 treat both failure types under the failure-aware policy: they receive zero. Without this policy—if invalid rows were silently excluded—Gemini's Telugu BGE score would rise from 0.630 to approximately 0.82, indistinguishable from GPT-4o's 0.794. The failure-aware denominator is therefore not cosmetic: it materially changes model ranking.

Why semantic metrics miss failures. A direct test of this blind spot is how automatic scores relate to English word overlap. On the Ollama outputs, raw BERTScore-F1 correlates with English word-overlap at $r = 0.955$ and BGE-M3 at $r = 0.805$: both metrics largely reward how much of the output remains English rather than how well it is romanized. Consistent with this, Ollama's verbatim-English Telugu responses receive BGE = 1.000 and BERTScore-F1 = 1.000, while its genuine romanized Telugu averages only 0.67—the metric scores a complete task failure higher than a success. Raw BERTScore-F1 is further compressed by a high floor: empty responses still score ≈0.42, because baseline rescaling is disabled and multilingual-BERT embeddings occupy a narrow cone. These properties explain why the metric assigns its highest scores to its worst failures.

4.3 Human Evaluation

Automatic metrics operate on surface similarity. BGE-M3 computes cosine distance between embedding vectors; BERTScore compares contextual token representations. Both measure whether a romanized output is semantically close to the English reference—but neither can answer the question that actually matters: does this read like something a native speaker would type?

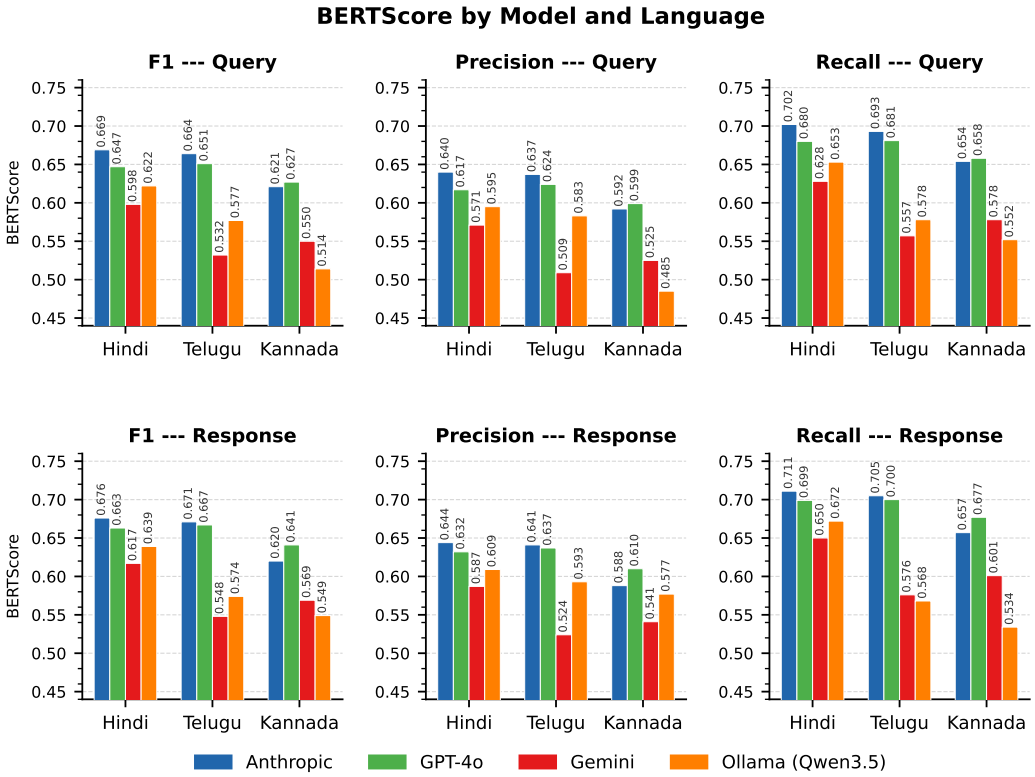


Figure 2: Multilingual BERTScore (F1, Precision, Recall) for romanized queries (top row) and responses (bottom row) across all four models and three languages. Scores use bert-base-multilingual-cased and are failure-aware (invalid rows receive zero).

Our automatic scores make the problem concrete. Across the two strongest models, Anthropic and GPT-4o, BGE-M3 response scores span 0.733–0.837 across all language conditions—a band that, considered in isolation, appears acceptable. Yet automatic metrics cannot detect that a response uses the wrong register, chooses a word that technically exists in the language but sounds unnatural, or follows English sentence structure while using romanized vocabulary. A model that produces grammatically correct but robotic-sounding Hinglish receives the same score as one that sounds like a real person typing on their phone.

Human evaluation targets precisely what BGE and BERTScore cannot see. Native speakers of Hindi, Telugu, and Kannada rated approximately 24 responses per language (73 total) across all four models on five dimensions: naturalness and fluency, grammatical correctness, word choice accuracy, register appropriateness, and overall usability (each on a 2–3 point scale; maximum 12 points per response). Rows were selected to include metric-disagreement cases (where BGE and BERTScore diverge by more than 0.15 points), hard narrative prompts, and failure anchors—precisely the cases where automatic metrics are most likely to miss real quality differences.

Annotation procedure. To accelerate review without sacrificing quality, we first generated reference scores for all ten parameters using Claude Opus 4.8 as an LLM judge. Native speakers then received pre-filled sheets and were instructed to correct any value they disagreed with, reading each romanized_response before accepting or overriding the draft score. The final H6–H10 scores are native-speaker judgments where a reviewer has corrected the draft; uncorrected rows remain at the LLM-judge value. To gauge whether the LLM judge can stand in for native speakers, a native reader re-scored a blind sample of Telugu rows. Agreement is poor: although the mean signed difference is near zero, this masks a mean absolute deviation of roughly 4 points out of 12, because the judge errs in opposite directions

Model	Nat. /3	Gram. /2	Word /2	Reg. /2	Usab. /3	Total /12	BGE (ref)
GPT-4o	2.49	1.93	1.97	1.94	2.40	10.73	0.790
Anthropic	2.05	1.66	1.70	1.73	2.15	9.30	0.786
Gemini	1.89	1.73	1.54	1.70	2.06	8.91	0.686
Ollama	0.76	0.89	0.93	1.04	0.85	4.48	0.641

Table 3: Human-corrected LLM-judge scores (weighted mean across all languages, 73 total rows: ∼24 per language). Telugu and Kannada rows have native-speaker corrections applied over the LLM-judge draft; Hindi scores are LLM-judge reference values (provisional, pending native-speaker review). Nat.=Naturalness/Fluency, Gram.=Grammar Correctness, Word=Word Choice Accuracy, Reg.=Register Appropriateness, Usab.=Overall Usability. BGE column shows mean failure-aware BGE-M3 response score from Table 1 for reference.

that cancel. It over-rates genuine failures—one Gemini Telugu response containing native-script characters and a stray Russian word received 10.4/12 from the judge but 3/12 from the native reader—and under-rates legitimate English-task outputs, which it penalizes for not being in the target language. The LLM judge is therefore subject to the same blindness as the embedding metrics, and we treat native judgments, not the judge, as ground truth. Hindi scores are LLM-judge reference values pending native-speaker review and should be treated as provisional. Telugu and Kannada scores reflect native-speaker corrections applied over the LLM-judge draft. Whether an LLM judge can reliably substitute for native speakers on romanized Indic evaluation is itself a finding we return to in Section 5.

Metric–human disagreement. Table 3 exposes the central finding of this paper. BGE-M3 scores for Anthropic (0.786) and GPT-4o (0.790) are nearly indistinguishable—a difference of 0.004 points that no downstream decision would act on. Native speakers disagree sharply: GPT-4o scores 10.73 out of 12 while Anthropic scores 9.30, a gap of 1.43 points driven primarily by naturalness (2.49 vs 2.05) and register appropriateness (1.94 vs 1.73). The metric ranking and the human ranking disagree at the top.

The gap is most severe for Ollama. BGE-M3 assigns it 0.641—poor, but not catastrophic by the metric's scale. Human raters assign a mean of 4.48 out of 12, with Telugu and Kannada responses averaging 4.07 and 2.86 respectively (largely unusable), while Hindi averages 8.28—a divergence explained by Ollama's near-complete instruction collapse in Dravidian languages (30% of Kannada outputs returned in English) versus better Hindi coverage in its training data. BGE cannot distinguish between a response that is semantically adjacent but linguistically broken and one that is simply the English source returned verbatim—both score near 1.000 under that policy. Human raters make this distinction immediately.

Language and register effects. Human scores follow the same language ordering as automatic metrics: Hindi obtains the highest mean (9.42/12, provisional—LLM-judge scores pending native-speaker review), followed by Kannada (7.11/12) and Telugu (6.89/12). The Hindi figure should be treated with caution as it has not yet been corrected by a native speaker, unlike the Telugu and Kannada scores which reflect direct native-speaker judgments. Register appropriateness is the most commonly flagged failure dimension in evaluator notes: Gemini responses in all three languages were described as "too formal" or using "very formal words" inconsistent with the conversational register of the prompts. One GPT-4o Kannada response was noted as "almost English"—a high-overlap output that BGE would score near 1.0 but that native speakers identified as a task failure.

Overall comparison. Table 4 consolidates automatic and human evaluation results across all models. The contrast between automatic and human ranking is the central finding: BGE-M3 separates Anthropic and GPT-4o by only 0.004 points, while human raters assign them a 1.43-point gap. Ollama receives a BGE score of 0.641—below the other models but still within a plausible range—yet human raters score it 4.48 out of 12, less than half the score of any other model.

Model	BGE-M3 (mean)	BERT-F1 (mean)	Human /12 (mean)	English-only failures
GPT-4o	0.790	0.657	10.73	0
Anthropic	0.786	0.656	9.30	2
Gemini	0.686	0.578	8.91	2
Ollama	0.641	0.587	4.48	41

Table 4: Summary of all evaluation results. BGE-M3 and BERT-F1 are failure-aware means over all 142 responses per model (averaged across languages). Human /12 is the weighted mean human score across 73 evaluated rows (∼24 per language; Hindi provisional). English-only failures are genuine cases (excluding English-task prompts) where the model returned English instead of the romanized target language, counted over the full ∼142-response set. GPT-4o is the only model with zero English-only failures across all three languages.

5 Discussion

Automated semantic evaluation can mask failures. BGE-M3 scores for the two strongest models (Anthropic, GPT-4o) cluster in the 0.73–0.84 range across all twelve model–language conditions. Considered in isolation, this band suggests adequate semantic preservation. Our analysis reveals three systematic failure modes that this band conceals. First, outputs that return the English source verbatim—a complete task failure—receive BGE scores of 1.000, because the metric measures English-to-English similarity rather than romanization quality. Thirty-five rows across Ollama Telugu and Kannada fall into this category (16 and 19 respectively, matching the failure counts in Table 2). Second, Gemini responses containing full paragraphs of native Telugu or Kannada script receive BGE scores of 0.80–0.84—passing marks for the wrong script entirely. Third, BGE and BERTScore disagree by more than 0.15 points on 44% of all 1,704 evaluated rows (maximum gap: 0.28 on a single response), indicating that the two metrics are not measuring the same underlying property. When two standard metrics diverge this substantially, neither can be trusted as a sole arbiter of quality.

Model-specific failure modes. Gemini and Ollama fail in qualitatively different ways that reflect different aspects of instruction following. Gemini produces script corruption: it generates otherwise fluent romanized text but switches to native script for certain word classes, particularly abstract nouns and technical terms. Frequency increases with script complexity (5.6% for Hindi, 16.9% for Telugu). Ollama produces instruction collapse: for Dravidian languages it returns the English source unchanged, suggesting the model has insufficient exposure to romanized Telugu and Kannada to complete the task. Crucially, these failures do not co-occur: Gemini never copies English and Ollama never leaks native script. The failure-aware scoring policy is essential for exposing both, because silently excluding invalid rows would raise Gemini's Telugu BGE from 0.630 to approximately 0.82—indistinguishable from GPT-4o's 0.794.

Naturalness beyond embedding similarity. Independent analysis of the romanized outputs reveals a consistent register problem that no automatic metric captures. All three API models generate text that follows English syntactic structure with Indic words inserted at corresponding positions—a pattern native speakers immediately identify as translated rather than composed. Gemini Hindi consistently uses hyper-formal Sanskrit-register vocabulary (saanskritik mahatva, adhikar hota hai) that no speaker would produce in informal digital communication. GPT-4o Kannada uses language-specific affirmatives (Haudu, Khandita) that signal greater lexical competence, but collapses into the same translated-English architecture for expository prose. Among ollama's successful Telugu outputs—those that are neither verbatim English nor repetition loops—we observe authentic discourse particles (raa, Aa, Aye) absent from all other model–language combinations, indicating real romanized-Telugu competence when the model engages the task. That competence does not survive aggregation: 16 of ollama's Telugu responses are verbatim English and several others degenerate, so its mean human score is 4.07/12 (its worst language), while ollama's strongest

Table 5: Mean output-token usage by model and target language. Counts include both generated fields and structured-output syntax.

Model	Hindi	Telugu	Kannada	Overall
Anthropic	1,059	1,037	1,107	1,068
GPT-4o	714	691	737	714
Gemini	633	686	717	678
Qwen3.5-35B	714	447	1,768	976

language is in fact Hindi (8.28/12). For the stronger models, the shared BGE range of 0.73–0.84 masks these divergences entirely.

Query-type effects. Pooling scores across all models and languages, emotional and narrative queries obtain significantly lower BGE scores (mean: 0.719) than technical and scientific queries (mean: 0.816), a gap of 0.097 points (t-test $p < 0.0001$). This is counterintuitive: one might expect code or mathematics to be harder to romanize than conversational text. The explanation is that technical terms have stable, widely-accepted romanizations (e.g., algorithm, hybridization) while emotional and idiomatic language resists literal transliteration. Prompts requiring extended first-person narrative or complex persona—such as roleplay and creative writing—are the hardest class for all four models across all three languages. This finding suggests that evaluation datasets for romanized generation should stratify by query type rather than treating all prompts as equivalent.

Output-token efficiency. Petrov et al. Petrov et al. (2023) showed that non-English input requires disproportionately more tokens under English-dominant tokenizers—a cost asymmetry that disadvantages non-English users before the model runs. Our data reveal a parallel asymmetry on the output side. Table 5 reports the mean model-reported output tokens for each paired query–response generation.

Anthropic reports the highest stable token usage, averaging 1,068 tokens, compared with 714 for GPT-4o and 678 for Gemini. This difference should not be interpreted directly as additional content or cost because each provider uses a different tokenizer. GPT-4o is the most consistent across languages, while Gemini reports the lowest overall mean. Qwen3.5-35B is substantially less stable: its low Telugu mean is partly caused by incomplete or zero-token outputs, whereas its Kannada mean is inflated by 49 generations that reached the 4,096-token limit, frequently through repetition. Thus, mean token usage is informative for the stable commercial systems but must be interpreted together with completion and degeneration rates for Qwen.

Implications for multilingual evaluation. Romanized Indic generation requires a metric bundle rather than any single score: semantic preservation (BGE-M3 or BERTScore), coverage rate (fraction of valid outputs), script compliance (native-character count), repetition rate, and native-speaker acceptability. The failure-scoring policy must be stated explicitly, because the standard practice of excluding failed outputs can change model rankings entirely. More broadly, the results suggest that embedding-based cross-lingual metrics are insufficient to evaluate generation quality for low-resource romanized varieties—a population of hundreds of millions of users currently invisible to standard NLP benchmarks.

6 Limitations

Source data and task design. The 142 source pairs are drawn from English OASST2, so the experiment measures controlled translation and romanization rather than naturalistic response to real romanized user input. Users who type in romanized script produce organically code-mixed, spelling-variable queries that differ systematically from the formal English prompts used here. Results may underestimate real-world quality degradation for models that struggle with informal romanized input.

Single prompting strategy. All four models receive the same structured JSON prompt. Different prompting strategies—chain-of-thought, few-shot examples, native-script interme-

diate steps—may substantially change output quality, particularly for Ollama's Kannada and Telugu failures. The results characterize one prompting regime, not the ceiling of each model's romanization capability.

Automatic metrics. BGE-M3 cosine similarity is our primary metric, but it assigns passing scores to verbatim English copies (BGE = 1.000) and native-script outputs (BGE ≈ 0.80–0.84), demonstrating its own limitations for this task. Raw BERTScore F1 using bert-base-multilingual-cased is difficult to interpret for cross-lingual English-to-romanized comparison, as its token alignment was not designed for this setting. Neither metric captures naturalness, register, or spelling consistency.

Human evaluation coverage. Human ratings are collected on a purposively sampled subset of approximately 24 responses per language (73 total) rather than the full 142-row dataset. Evaluators are sampled from a single team; ratings from speakers across different regions, age groups, and levels of digital literacy may differ. Romanization has no single universally correct spelling, and rater agreement on naturalness judgments was not formally measured. The human evaluation results should be interpreted as directional evidence rather than a definitive quality ranking.

Languages and models. We study Hindi, Telugu, and Kannada. Other Indic romanized varieties—Bengali, Tamil, Marathi—may exhibit different patterns. The four models evaluated here (Claude, GPT-4o, Gemini, Qwen3.5-35B) represent a snapshot of current capabilities; results for other model families or future versions may differ.

7 Conclusion

We evaluated four LLMs—Claude, GPT-4o, Gemini, and Qwen3.5-35B—on romanized Hindi, Telugu, and Kannada generation under a failure-aware scoring policy that assigns zero to empty, copied, and script-corrupted outputs. Three findings stand out. First, language is a stronger predictor of output quality than model choice: all four models follow the same Hindi > Telugu > Kannada ordering, consistent with the relative availability of romanized Indic text in pretraining data. Second, Gemini and Ollama exhibit non-overlapping failure modes—script corruption and instruction collapse respectively—that are invisible without explicit coverage tracking, and that failure-aware scoring exposes by materially changing model rankings. Third, and most importantly, BGE-M3 cosine similarity assigns scores in the 0.73–0.84 range across all language conditions for the two strongest models (Anthropic and GPT-4o), including outputs that return pure English verbatim (BGE = 1.000) and outputs rendered entirely in native script (BGE ≈ 0.82). Native-speaker human evaluation reveals systematic quality failures—unnatural register, wrong word choice, inconsistent romanization— that this band conceals. The metric cannot see the gap that users experience.

Romanized generation for Indic languages requires evaluation frameworks that combine semantic metrics with coverage, script compliance, and native-speaker judgment. We release our scoring pipeline, the failure-aware evaluation protocol, and the human annotation rubric to support future benchmarking of this underserved but large-scale user population.

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Reproducibility Statement

Data. Source prompts are drawn from the OpenAssistant dataset (OASST2; Köpf et al. 2023). We use 142 question–answer pairs per language, selected by filtering for English-language assistant turns and stratifying by query type (technical, emotional, narrative, factual). The full input manifest with OASST2 message IDs will be released with the artifact.

Models. We evaluate four systems: Anthropic Claude Opus 4.8 (claude-opus-4-8), OpenAI GPT-4o (gpt-4o-2024-11-20), Google Gemini (gemini-2.0-flash), and Qwen3.5-35B served locally via Ollama (ollama/qwen3.5:35b). All models were queried with a single prompt template instructing the model to produce a romanized response in the target language, using default sampling settings unless otherwise specified. The complete prompt is included in the artifact.

Metrics. BGE-M3 cosine similarity is computed with the BAAI/bge-m3 checkpoint via the FlagEmbedding library (v1.2). BERTScore F1 uses bert-base-multilingual-cased with the HuggingFace evaluate library (v0.4). Both metrics are applied under the failure-aware policy: rows with empty output, native-script corruption, or English verbatim copy receive zero and are included in the denominator.

Human evaluation. Approximately 24 rows per language (73 total: 24 Telugu, 24 Kannada, 25 Hindi) were scored on five dimensions (maximum 12 points). Telugu and Kannada rows

received native-speaker corrections applied over LLM-judge pre-fills; Hindi scores are LLM-judge reference values pending native-speaker review. Rubric definitions and the LLM-judge pre-fill methodology are detailed in the supplementary scoring guide, which will be released with the artifact. Inter-annotator agreement was not formally measured; each language was reviewed by a single native speaker.

Code. The generation pipeline, evaluation scripts, plotting code, and human evaluation rubric will be released as an anonymous artifact prior to final submission. A README with exact reproduction commands will be included.

Use of Language Models

Language models were used in the research workflow for data generation, evaluation, code assistance, and drafting assistance. The authors reviewed all generated text, code, and claims and remain responsible for the contents of the paper.